ALOKNATH SAH

Senior Software Engineer | React.js | TypeScript | Frontend Architecture | Performance | Micro Frontends

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Summary

Senior Software Engineer (Frontend Heavy) with 5+ years of experience building **scalable**, high-performance web applications using React.js, TypeScript and JavaScript. Strong expertise in **frontend architecture**, **micro-frontends, performance engineering, virtualization, client-side caching, accessibility** and reusable component systems. Experienced in building data-intensive applications from the ground up, optimizing browser rendering and **mentoring** engineers. Additional experience with **Next.js**, **Node.js**, REST APIs, Azure and **GenAI**-powered applications.

Technical Skills

- **Frontend**: React.js, React Hooks, TypeScript, Next.js, JavaScript, HTML5, CSS3, Tailwind CSS, Material UI.
- **Frontend Architecture:** Component Architecture, Design Systems, Micro Frontends, Webpack Module Federation, Single-SPA, Vite, npm, Reusable Components.
- **Performance:** React Performance, Virtualization, Lazy Loading, Code Splitting, Infinite Scrolling, Browser/DOM Optimization, Lighthouse.
- **State & Data:** Redux, Redux Toolkit, Context API, IndexedDB, WebSockets, Client-Side Caching.
- **Accessibility & Testing:** WCAG, ARIA, Accessibility, Jest, cypress, React Testing Library.
- **Backend**: Node.js, Python, FastAPI, REST APIs, MongoDB, SQL Server.
- **AI:** LLM-integrated applications, GenAI workflows**.**

Work Experience

Affine | Bangalore, India
Senior Software Engineer **July 2025 - Present**

- **Architected** and delivered **scalable** React.js and TypeScript applications from the ground up, **owning frontend architecture**, component design, API integration, performance engineering, and production delivery.
- **Built a GenAI-powered React application from the ground up for Intertek**, automating complex manual report-generation workflows and reducing report preparation time from approximately **2 days to ~15 minutes**.
- Led frontend development and mentored **2 developers**, driving architecture reviews, modular component design, code consistency, and production-quality standards.
- Conducted **Lighthouse audits** and optimized render-blocking resources and bundle sizes, improving **Lighthouse performance** score.
- Developed frontend experiences from scratch with optimized rendering for **large datasets**, using lazy loading and infinite scrolling to improve application responsiveness.
- Designed and implemented a **client-side caching** layer using **IndexedDB**, reducing redundant network requests and improving responsiveness for frequently accessed application data.
- **Optimized a data-intensive React application responsible for rendering 100+ pages**, identifying browser rendering, DOM, and memory bottlenecks and redesigning the rendering strategy to maintain browser stability at scale.
- **Implemented virtualization for large and DOM-intensive interfaces**, ensuring that only visible content was rendered and preventing browser freezes caused by excessive DOM nodes and rendering overhead.
- Improved application responsiveness through optimized **infinite scrolling**, **IndexedDB caching**, and React re-render reduction techniques.

Sparkviz Technologies | Bangalore, India
Frontend Developer **June 2023 – July 2025**

- Developed a **multi-step clinical assessment** and risk analytics platform for a US-based healthcare client, featuring dynamic questionnaires and interactive clinician dashboards.
- Engineered persistent form hydration using **IndexedDB caching**, preventing browser memory bottlenecks on deep form trees and enabling users to seamlessly resume long assessments from their exact drop-off point upon return or page refresh.
- Designed reusable React components and shared frontend patterns to improve maintainability, reduce duplicated implementation, and provide consistent UI behavior across application modules.
- Delivered a high-density **clinical dashboard** to visualize **AI-generated risk scores**, breakdown percentiles, and longitudinal trends with interactive charts.

- Implemented strict **WCAG AA** accessibility standards, incorporating comprehensive **screen-reader**, keyboard-only focus traps for modals, and semantic form step indicators.
- Applied Low-Level Design (**LLD**) principles across both platforms to standardize modular form controls, UI state managers, and **RBAC** security patterns, cutting cross-module code duplication.
- **Improved page load time** by 30% through **bundle optimization**, lazy loading and rendering improvements.
- Architected and owned the **Fleet Onboarding & Client Freight Booking Micro Frontend** using **React.js, TypeScript, and Webpack Module Federation**, enabling zero-downtime, independent deployments
- Implemented **list virtualization** using React Window for 2,000+ fleet records, reducing DOM overhead and maintaining **responsive scrolling**.
- Contributed to a TypeScript/React **monorepo** with shared packages and reusable UI components, establishing dependency boundaries and **shared frontend infrastructure** across applications.

Ajio.com (Reliance Retail) | Bangalore, India
SDE 1 | React Developer

Jul 2021 - May 2023

- Led UI/UX enhancements on order pages, reducing drop-offs by **18%**.
- Developed a user-friendly **Rating feature** in the UI, allowing customers to evaluate their purchased products.
- Developed customer rating system & gamification flows decreased drop off rates by **6%.**
- Implemented **Material UI theme** configurations and **reusable styling patterns** to maintain consistent visual behavior across application modules.
- Collaborated with product managers and designers to **translate UI requirements** and prototypes into responsive, accessible React interfaces.
- Improved **email templates in the notification engine**, ensuring better user engagement.
- Enhanced **Google Analytics (GA)** instrumentation across key user journeys, implementing and improving frontend event tracking to capture user interactions and behavioral patterns, enabling product teams to monitor engagement and make data-driven decisions.
- **Contributed to A/B testing (experimentation) initiatives**, enabling different UI experiences and feature variants to be served to different user segments and supporting data-driven evaluation of feature and product behavior.

Awards

- **Game Changer of the month Award – 3X**

Education and Training

Full-Stack Web Development Program. **Oct 2019 - Jun 2021**
Masai School | Bangalore, India
Intensive, immersive program focused on Full-Stack JavaScript, Data Structures & Algorithms, React.js, and Modern Web Architecture. Completed 1,200+ hours of hands-on coding and collaborative team projects.

Bachelor of Technology. **May 2015 - Jun 2019**
Indira Gandhi Institute of Technology, Sarang